

Question 1

Not yet answered

Marked out of 1.00

What is the purpose of regularization in machine learning?

- ☐ a. To prevent overfitting by penalizing model complexity
- ☐ b. To increase model complexity
- ☐ c. To speed up training
- ☐ d. To visualize results

Question 2

Not yet answered

Marked out of 1.00

What is the bias-variance tradeoff in machine learning?

- ☐ a. Balance between training time and accuracy
- ☐ b. Balance between model complexity and performance
- ☐ c. Balance between data size and model size
- ☐ d. Balance between supervised and unsupervised learning

Question 3

Not yet answered

Marked out of 1.00

What is the difference between bagging and boosting ensemble methods?

- ☐ a. Both train models sequentially
- ☐ b. No difference
- ☐ c. Bagging trains models sequentially, boosting trains in parallel
- ☐ d. Bagging trains models in parallel, boosting trains sequentially

Question 4

Not yet answered

Marked out of 1.00

In decision trees, what is information gain?

- ☐ a. The depth of the tree
- ☐ b. The reduction in entropy after splitting on an attribute
- ☐ c. The amount of data processed
- ☐ d. The number of leaf nodes

Question 5

Not yet answered

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What is the main difference between supervised and unsupervised learning?

- ☐ a. Supervised learning uses labeled data, unsupervised learning uses unlabeled data
- ☐ b. Both use the same type of data
- ☐ c. Supervised learning is faster
- ☐ d. Unsupervised learning uses labeled data, supervised learning uses unlabeled data

Question 6

Not yet answered

Marked out of 1.00

What is feature engineering?

- ☐ a. Removing features from data
- ☐ b. Sorting features alphabetically
- ☐ c. Creating new features from existing data
- ☐ d. Converting features to numbers

Question 7

Not yet answered

Marked out of 1.00

In a confusion matrix for binary classification, what does precision represent?

- ☐ a. $(\text{True Positives} + \text{True Negatives}) / \text{Total}$
- ☐ b. $\text{True Positives} / (\text{True Positives} + \text{False Positives})$
- ☐ c. $\text{True Positives} / (\text{True Positives} + \text{False Negatives})$
- ☐ d. $\text{True Negatives} / (\text{True Negatives} + \text{False Positives})$

Question 8

Not yet answered

Marked out of 1.00

What is ensemble learning?

- ☐ a. Using only neural networks
- ☐ b. Combining multiple models for better performance
- ☐ c. Learning one model at a time
- ☐ d. Using only decision trees

Question 9

Not yet answered

Marked out of 1.00

What is the difference between parametric and non-parametric algorithms?

- ☐ a. Parametric algorithms are always better
- ☐ b. No difference
- ☐ c. Parametric algorithms have fixed number of parameters, non-parametric don't
- ☐ d. Non-parametric algorithms have fixed number of parameters, parametric don't

Question 10

Not yet answered

Marked out of 1.00

What is overfitting in machine learning?

- ☐ a. When a model uses too much memory
- ☐ b. When a model performs well on training data but poorly on test data
- ☐ c. When a model takes too long to train
- ☐ d. When a model performs poorly on both training and test data

Question 11

Not yet answered

Marked out of 1.00

What is the difference between L1 and L2 regularization?

- ☐ a. No difference
- ☐ b. L1 is faster than L2
- ☐ c. L2 creates sparse models, L1 shrinks coefficients evenly
- ☐ d. L1 creates sparse models, L2 shrinks coefficients evenly

Question 12

Not yet answered

Marked out of 1.00

Which evaluation metric is most appropriate for imbalanced datasets?

- ☐ a. Mean Squared Error
- ☐ b. F1-score
- ☐ c. Accuracy
- ☐ d. R-squared

Question 13

Not yet answered

Marked out of 1.00

What is reinforcement learning?

- ☐ a. Learning with labeled data
- ☐ b. Learning with unlabeled data
- ☐ c. Learning through interaction with environment using rewards
- ☐ d. Learning by memorizing examples

Question 14

Not yet answered

Marked out of 1.00

What is the purpose of cross-validation in machine learning?

- ☐ a. To visualize data
- ☐ b. To reduce data size
- ☐ c. To evaluate model performance on multiple train-test splits.
- ☐ d. To increase training speed

Question 15

Not yet answered

Marked out of 1.00

What is the purpose of Principal Component Analysis (PCA)?

- ☐ a. Dimensionality reduction
- ☐ b. Clustering
- ☐ c. Regression
- ☐ d. Classification

Question 16

Not yet answered

Marked out of 1.00

What is the curse of dimensionality?

- ☐ a. Having missing values
- ☐ b. Having too few features
- ☐ c. Having too many features leading to sparsity and increased complexity
- ☐ d. Having categorical features

Question 17

Not yet answered

Marked out of 1.00

In K-means clustering, how do you determine the optimal number of clusters?

- ☐ a. Use the number of features
- ☐ b. Use the elbow method or silhouette analysis
- ☐ c. Use random selection
- ☐ d. Always use 3 clusters

Question 18

Not yet answered

Marked out of 1.00

What is the difference between classification and regression problems?

- ☐ a. No difference
- ☐ b. Classification predicts continuous values, regression predicts categories
- ☐ c. Classification is always more accurate
- ☐ d. Classification predicts categories, regression predicts continuous values

Question 19

Not yet answered

Marked out of 1.00

Which algorithm is best suited for binary classification problems?

- ☐ a. Principal Component Analysis
- ☐ b. K-means clustering
- ☐ c. Linear regression
- ☐ d. Logistic regression

Question 20

Not yet answered

Marked out of 1.00

How does the Support Vector Machine (SVM) algorithm work?

- ☐ a. By finding the optimal hyperplane that maximally separates classes
- ☐ b. By using decision rules
- ☐ c. By clustering similar data points
- ☐ d. By finding the mean of data points